Day: M T W T F S   Date: 27/12/2021

Linear Algebra

Muhammad Hassan Khalid

211-5692

(2) Equation for the image of line $y = -4x + 3$; under multiplication by the matrix

$$A = \begin{bmatrix} 4 & -3 \\ 3 & -2 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 3 & -2 \end{bmatrix}^{-1} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2 & 3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$x = -2x' + 3y' \quad -(1)$$
$$y = -3x' + 4y' \quad -(2)$$

Substituting (1) and (2) values in the line equation

$$y = -4x + 3$$
$$-3x' + 4y' = -4(-2x' + 3y') + 3$$
$$-3x' + 4y' = +8x' - 12y' + 3$$
$$4y' + 12y' = 8x' + 3x' + 3$$
$$16y' = 11x' + 3$$
$$\boxed{y' = \frac{11}{16}x' + \frac{3}{16}} \longrightarrow \text{Ans}$$